Microinverter Upgrade Workflow and Division of Labor

Doc. No.	_____	Version	V1.0
Effective date	2026-11-13	Scope	Microinverter firmware upgrade line

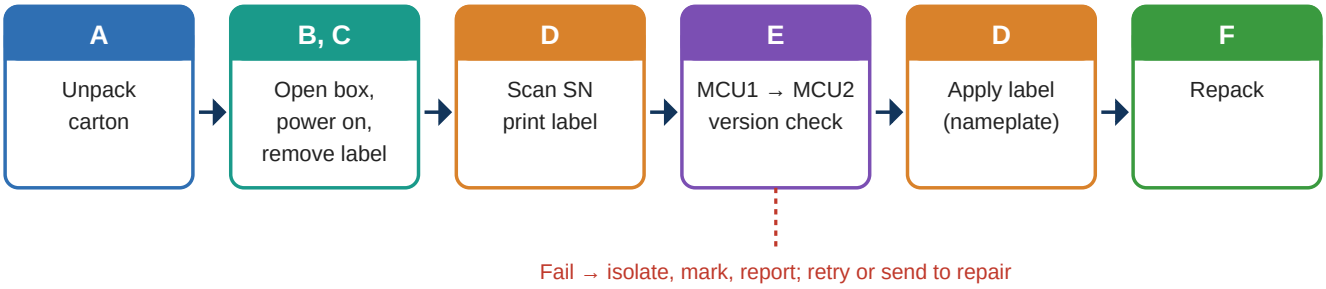
1. Purpose and Scope

This document standardizes the batch firmware upgrade of microinverters. It defines who does what, the key operating points and quality controls, so that results are traceable, labels match devices one-to-one, batches are never mixed and no unit is missed. It applies to everyone working on the upgrade line.

2. Staffing (6 people: A–F)

Station	Staff	Steps	Responsibilities
A	1	Step 1 Unpack	Open the large cartons and hand them to D
B, C	2	Open small boxes, power on, Step 2 Remove labels	Open small boxes, take out the microinverters, connect power, peel off the old labels
D	1	Step 3 Scan SN, Step 4 Print labels, Step 7 Apply labels	Receive large cartons from A, scan and print labels; after a passed upgrade, stick the label on the nameplate
E	1	Step 5 Upgrade, Step 6 Check result	Run the upgrade program and confirm the result
F	1	Step 8 Repack	Put the labeled microinverters back into their original cartons

3. Process Overview



Step	Operation	Station	Input	Output
Step 1	Unpack (large carton)	A	Original large carton	Opened carton + small boxes
—	Open small boxes, power on	B, C	Small boxes	Powered microinverters
Step 2	Remove old labels	B, C	Units with old label	Units without old label
Step 3	Scan SN	D	Large carton / units	SN recorded in system
Step 4	Print labels	D	SN data	New labels
Step 5	Upgrade	E	Units to upgrade	Upgraded units
Step 6	Check result	E	Upgraded units	Pass / Fail
Step 7	Apply labels	D	Passed units + new labels	Labeled units
Step 8	Repack	F	Labeled units + carton	Finished carton

Flow: A → B, C → D (scan, print) → E (upgrade, check) → D (apply labels) → F (repack).

4. Operating Points by Station

A: Unpacking (Step 1)

- Open cartons batch by batch; process one batch at a time to avoid mixing.
- Check for damage, water ingress or deformation; set abnormal units aside and mark them.
- Hand the large carton to D and the small boxes to B and C; keep the large carton for final repacking.

B, C: Open small boxes, power on, remove labels (Step 2)

- Open the small boxes and take out the microinverters and accessories.
- Connect power as instructed and confirm the unit is powered normally.
- Peel off the old label without damaging the housing or nameplate area; remove adhesive residue.
- Collect old labels centrally to prevent SN confusion.
- Pass processed units to E in order.

D: Scan, print, apply labels (Steps 3, 4, 7)

- Receive the large cartons from A, scan each SN into the system/record sheet and confirm it matches the unit; report duplicates or unreadable codes immediately.
- Print new labels from the scanned SNs; check the first label for content and barcode readability; keep labels in device order.
- Collect misprinted or voided labels separately; never mix them with good labels.
- After a passed upgrade, verify the label SN matches the device SN, then stick the label on the nameplate – uniform position, flat, no bubbles. Only passed units get new labels.

E: Upgrade and result check (Steps 5, 6)

- Connect the device per the upgrade work instruction and run the upgrade with the specified firmware.
- Do not cut power, unplug or move the device during the upgrade.
- After both MCU1 and MCU2 are upgraded, run the version check program and confirm the firmware matches the target version.
- Passed units go to D for labeling; failed units are set aside, marked and reported to the person in charge, and must not go to the next step.
- Record results in the system/record sheet so SN and result are traceable.

! Upgrade Reminder

1	Upgrade **4 units** at a time.
2	Upgrade **MCU1** first. When finished, wait until the **red LED changes from blinking to steady** and the **blue LED turns off**, then start MCU2.
3	Same for **MCU2**: when finished, wait until the **red LED is steady** and the **blue LED is off**.
4	When both sides are done, run the **version check program**.

F: Repacking (Step 8)

- Put labeled units and accessories back into the matching original carton; check quantity, model and batch.
- Seal and stack the cartons; record the batch on the outside.
- Total the good and failed units of the day and reconcile with the input quantity.

5. Quality Control

1	**SN consistency**	scanned SN, label and device must match; verify before applying the label.
2	**No batch mixing**	one batch at a time, materials separated by batch at each station.
3	**Isolate failures**	units with failed upgrade or wrong version are stored and marked separately; never labeled or packed.
4	**Quantity balance**	input = good + failed + abnormal per batch; reconcile before sealing.
5	**Traceability**	keep SN, upgrade result, operator and time.

6. Exception Handling

Exception	Action
Damage / water ingress at unpacking	Set aside and mark, notify the person in charge, exclude from later steps
No response after power-on	Stop on that unit, mark and report
Abnormal LED state (red not steady / blue not off)	Do not start the next MCU; stop on that unit, mark and report
SN unreadable / duplicate	Stop on that unit, report for verification
Label misprint	Collect the voided label, reprint and recheck
Upgrade failure	Retry per rules; if it still fails, mark as defective and send to repair
Quantity mismatch	Stop repacking, trace back station by station, then release

7. Open Items

- ☐ Timing of D labeling: currently written as "after a passed upgrade"
- ☐ Firmware version, upgrade tool and work instruction version
- ☐ Maximum retries and repair process for failed units
- ☐ Who is notified about failed units
- ☐ Record sheet / system field requirements

Approval

Prepared by	Reviewed by	Approved by
Signature / Date	Signature / Date	Signature / Date